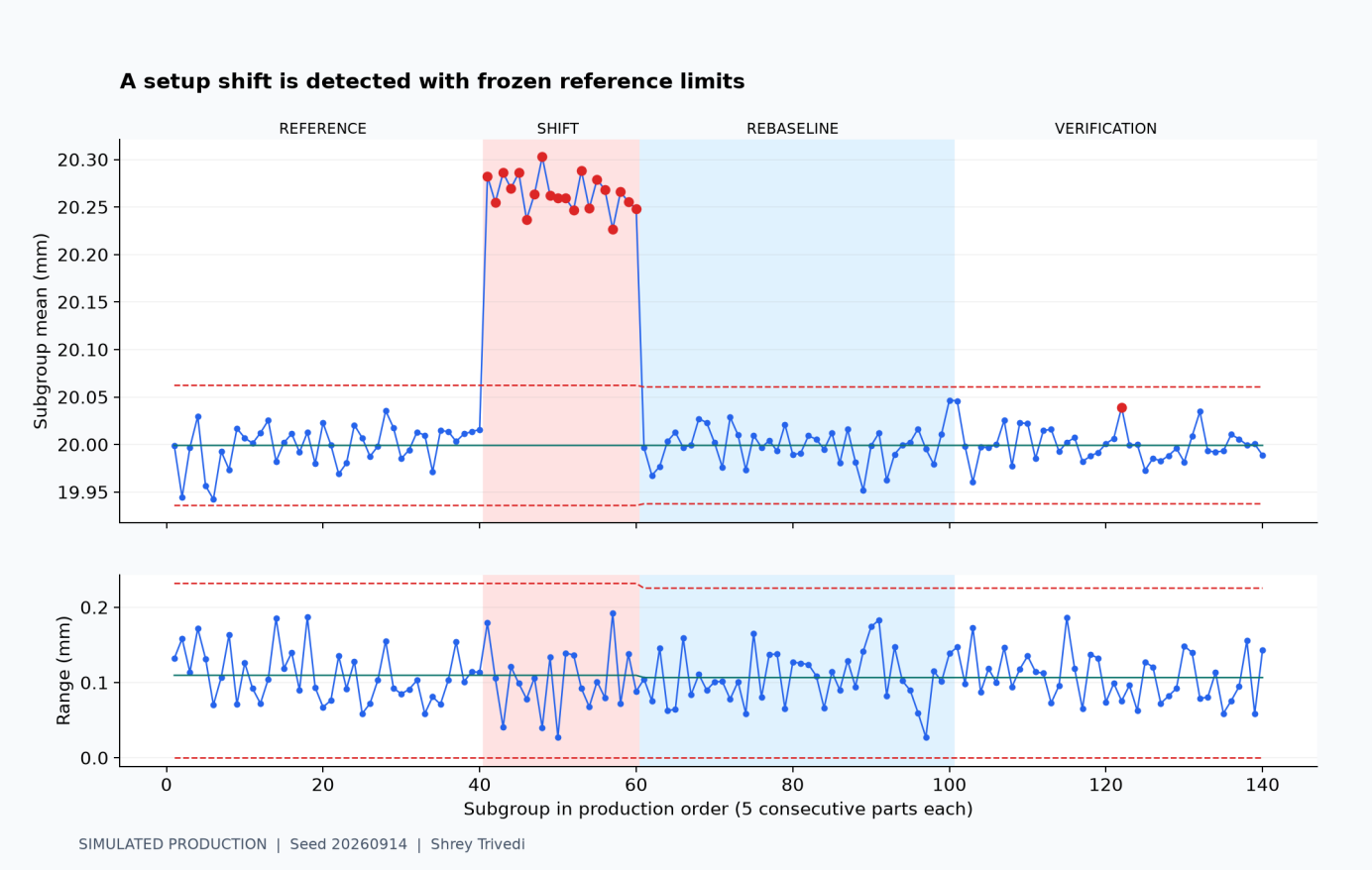


Manufacturing Quality Engineering Study

A connected simulation case: inspection integrity, measurement, SPC, corrective action and drawing-based inspection planning.

Shrey Trivedi | Portfolio edition | September 2026



Phase 1	Phase 2	Inspection exercise
20,000 part records 90 GR&R readings	700 ordered records 140 subgroups of five	7 characteristic IDs 1 first-piece record + 2 hole challenges

All production and inspection measurements are synthetic. Minitab execution is pending authenticated access. The later SPC verification contains a trend alert, so corrective-action effectiveness remains open. This packet reports that result without removing the alert.

01 / Case map

One characteristic ID follows the requirement through analysis and reaction.

Stock -> setup and gage check -> turning -> cross-drilling -> finishing -> first-piece / routine inspection -> disposition

D01 shaft diameter connects the Phase 1 inspection audit to the new ordered SPC challenge. H02 hole position is a separate inspection-planning example: the Phase 2 NCR does not claim hole-position measurements.

Requirement	Planning link	Evidence / reaction
D01: 19.70-20.30 mm	CP-D01 / IP-D01 / NCR-001	Frozen-limit alarm; challenge-lot hold; CA-001
L01: 49.50-50.50 mm	CP-L01 / IP-L01	Phase 1 length escapes repaired
H01/H02: hole size / position	CP-H01/H02 / IP-H01/H02	Bore size + full-axis/bonus example
E01/F01/S01: face / form / Ra	CP-E01/F01/S01	Individual method and first-piece record
G01: measurement adequacy	CP-G01	Simulated GR&R; physical validation required

Evidence boundaries

Phase 1 retains its original simulation and repaired inspection logic. Phase 2 is a separate time-ordered experiment. The Rev B drawing is an authored learning addendum to the original Onshape exercise. No physical CMM, shop-floor validation, actual approvals or formal AS9102 experience is asserted.

02 / Measurement and inspection

Keep the existing evidence and understand what it cannot establish.

Phase 1 result	Interpretation
1,067 / 20,000 fail; pass rate 94.665%	Independent diameter and length checks plus assigned defect labels.
75 net failures avoided in seed 42	Ideal centering leaves 992 failures; five former oversize labels become other primary defects.
GR&R SD 0.00976 mm; study variation 9.13%; ndc 15	10 parts x 3 operators x 3 trials. Mixed part range makes the global ratio look favorable.
Purpose-specific measurement assessment	Bias, linearity, stability, uncertainty and method suitability remain unverified physically.

Why the new SPC data are separate

The 20,000-row dataset did not establish rational subgroups or an actual production sequence. Sorting it by a date field would not repair that limitation. Phase 2 instead defines an explicit sequence, subgroup membership and generator truth.

Phase 2 measurement model

True process SD = 0.045 mm; independent measurement SD = 0.006 mm. Observed SD includes both. These are separate authored assumptions, not a transfer of the Phase 1 GR&R result. No measurement-noise subtraction is used for capability.

03 / SPC experiment design

700 observations; five consecutive parts per 15-minute interval.

Window	Groups / n	Design purpose
Reference	1-40 / 200	Estimate initial limits.
Setup challenge	41-60 / 100	Introduce +0.270 mm wrong diameter compensation. All are quarantined challenge data.
New baseline	61-100 / 200	Restore setup; estimate revised limits.
Verification	101-140 / 200	Apply frozen new limits to later observations.

Rules and estimates

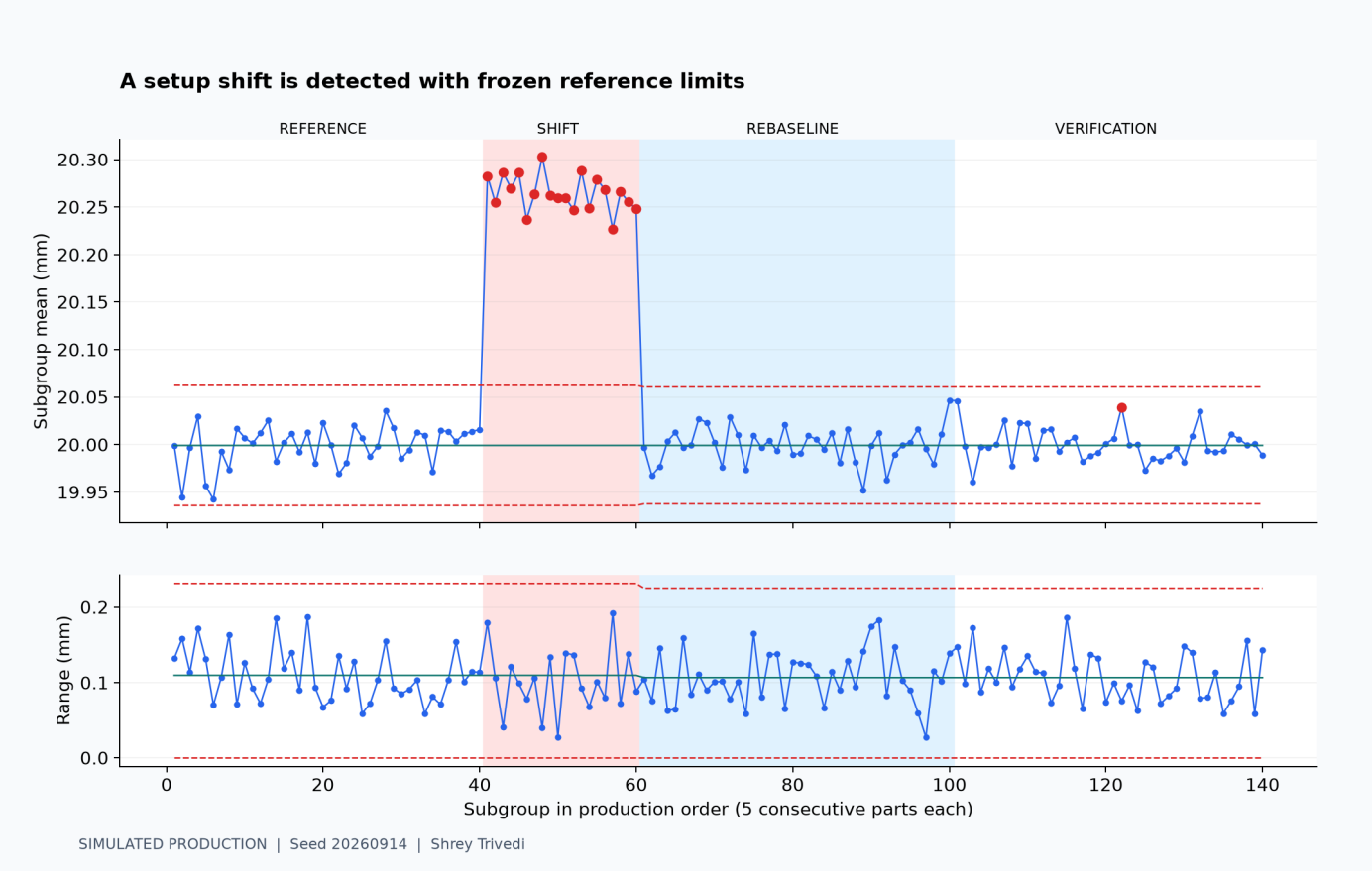
X-bar: point beyond 3 sigma (T1), nine on one side (T2), six monotonic points (T3). R: beyond-limit T1 only. Runs are marked at their final point. The chart uses $\bar{R}/d_2$ with $d_2=2.326$, $A_2=0.576802$, $D_3=0$ and $D_4=2.114$.

Five successive part times span four minutes. No unobserved parts are invented for the remaining interval. All windows and seed 20260914 were fixed before looking at output; no points or seeds were selected to create a clean result.

For a live process, the first signal triggers a stop/hold. The complete 20-subgroup challenge window is retrospective training evidence, not permission to keep making suspect parts.

04 / Control charts

The mean shifts while within-subgroup spread remains comparable.



Limits	X-bar center / LCL / UCL (mm)	R UCL (mm)
Reference	19.99904 / 19.93582 / 20.06227	0.23170
New baseline	19.99909 / 19.93759 / 20.06059	0.22539

All 20 shifted subgroup means signal against the reference limits. Reference and rebaseline have no selected-rule signals. The later verification has one T3 alert; inspect the alarm log and retain the flagged sequence. The model has no new assigned cause in that window, but that knowledge does not justify ignoring a real chart alarm.

05 / Capability with conditions

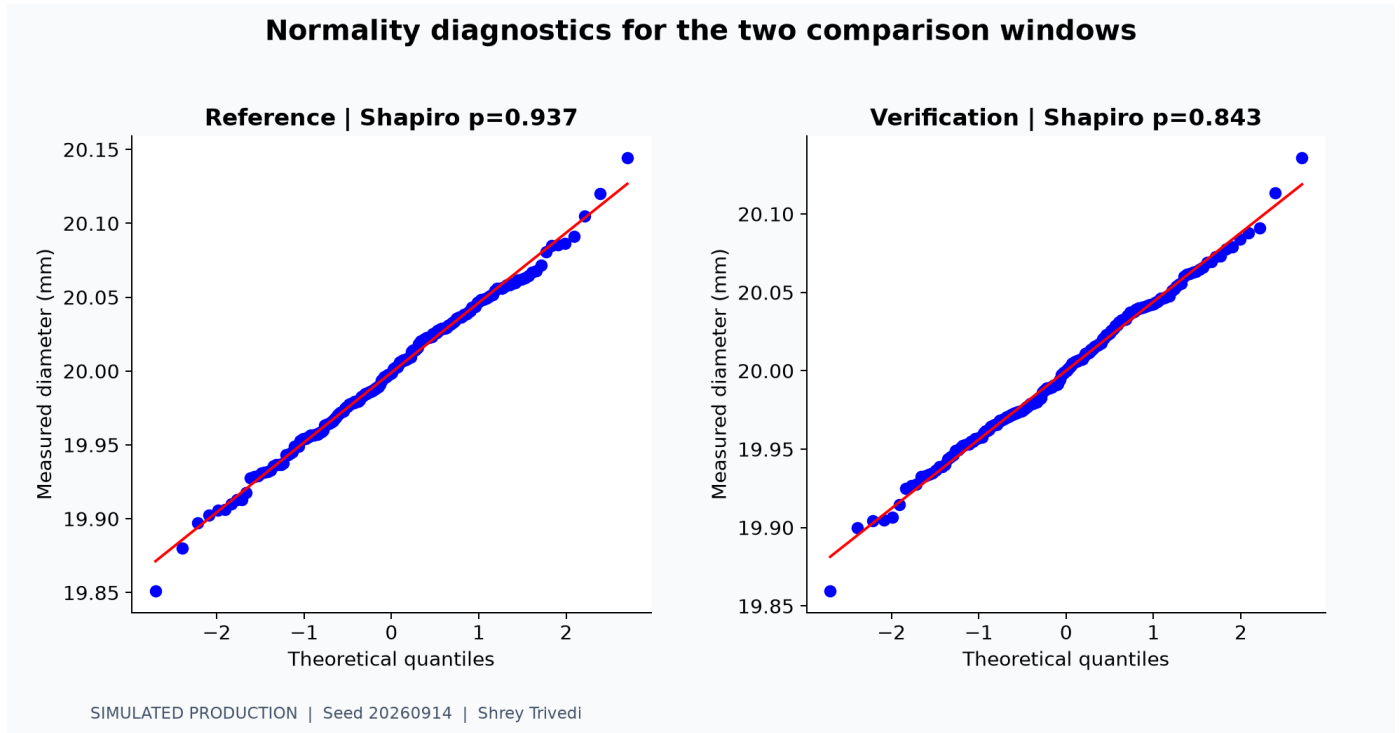
Specification limits describe acceptance. Control limits describe expected process behavior.



Reference and new-baseline estimates follow a no-signal finite-window review. Verification indices are provisional because T3 fired. Shifted indices are descriptive only. Pooled Ppk = 0.849 mixes distinct means and is not a stable capability claim.

06 / Distribution checks

Normality and independence are assessed, not established by a single p-value.



Reference Shapiro-Wilk $p=0.937$; verification $p=0.843$. Lag-1 correlations are -0.076 and -0.068. These diagnostics are consistent with the authored normal-noise model, but do not override chart signals.

Within SD is $R\bar{bar}/d2$; overall SD is the sample SD. Cp/Cpk use within variation, Pp/Ppk overall variation. Subgroup sampling noise can make Ppk exceed Cpk in this finite sample; it is not automatically an error.

The verification window has zero diameter failures in 200 parts. Under independent identical Bernoulli trials, its exact one-sided 95% upper bound is 1.487%. Zero observed failures is not a guarantee of zero future failures.

07 / NCR and containment

NCR-001 | D01 | Wrong diameter compensation after setup

Field	Simulated record
Detection	Subgroup 41, X-bar T1 using frozen reference limits.
Affected scope	SPC-041-1 through SPC-060-5; 100 quarantined challenge parts.
Nonconforming count	22 measured diameters exceed 20.300 mm.
Immediate reaction	Hold lot; preserve setup values and measurements; verify gage before adjustment.
Disposition	22 nonconforming pieces await engineering disposition; 78 dimensionally conforming pieces remain held pending review.
Responsibility	Proposed: setup technician, manufacturing engineer, quality engineer. No real signatures.

Causal evidence

The generator and Setup_Code prove that this model introduced a +0.270 mm diameter shift. Missing recipe comparison and independent first-piece review are hypothetical system-level causes used to design preventive controls. They are not interview findings.

Five Whys

High diameter -> wrong compensation -> manual entry without recipe comparison -> no independent first-piece gate -> no assigned release evidence after setup. The first two links are demonstrated by the simulation; the remaining links require real process records before acceptance in a factory.

08 / Root-cause review and action

Separate observed model evidence from alternative hypotheses.

Fishbone branch	Candidate cause	Evidence / next check
Method	Wrong compensation	Authored +0.270 shift / recipe audit
Machine	Tool/holder movement	Not modeled / physical inspection
Measurement	Gage zero bias	Noise unchanged in model / reference check
Material	Stock change	Not modeled / lot trace
People	Missing independent review	Hypothetical / sign-off audit
Environment	Temperature change	No Phase 2 thermal shift / collect temperatures

CA-001 actions

Restore approved setup at subgroup 61; add recipe comparison and first-piece checks to CP-D01; freeze new limits from groups 61-100; evaluate groups 101-140 separately. The restored model removes diameter rejects in the later windows.

Effectiveness is still open

The later run has 0/200 failures but one T3 alert. Review that six-point trend and define independent follow-up before real release. Do not reroll the seed, erase points or call this a completed 8D. This retained complication is part of the engineering conclusion.

09 / Datum and tolerance interpretation

Rev B learning addendum: A is the shaft OD axis; B is the seating end.

The primary cylindrical datum controls transverse translation and tilt. The secondary end face controls axial position. Rotation around the shaft is intentionally free because no cross-hole clocking requirement is assumed. A functional clocking requirement would need additional definition.

Callout	How to inspect / interpret
Basic 25 from B	The exact axial location of the true hole axis; position supplies the tolerance.
Position DIA0.10(M) A B	A cylindrical zone controlling full-axis location and orientation. At ideal bore size5.02, bonus0.07 gives allowed diameter0.17.
B face perpendicularity0.05 A	Two planes0.05 apart perpendicular to A; evaluate the face in that zone.
Cylindricity0.10	Two coaxial cylinders0.10 apart; form control, no datum.
Ra <=3.2 micrometres on OD	Profilometer with agreed cutoff/filter/evaluation length; not a visual defect label.

The original Onshape sheet is preserved. Rev B is a separate schematic inspection addendum with intentionally revised datum labels, not an edited native Onshape export. The original labels must not be mixed with this plan. The learning drawing is not released for manufacture.

10 / First-piece inspection record

FP-001 | All values are invented training measurements; no physical inspection performed.

ID	Specification	Example result	Decision
D01	19.70 to 20.30	20.014 mm	Pass
L01	49.50 to 50.50	50.06 mm	Pass
H01	4.95 to 5.05	5.02 mm	Pass
E01	<=0.05	0.018 mm	Pass
F01	<=0.10	0.041 mm	Pass
S01	<=3.2	2.4 um	Pass
H02	position <=0.17 at size5.02	0.09 mm	Pass

Hole position challenge records

Record	Actual / allowed zone diameter	Decision
FP-001	0.090 / 0.170 mm	Pass
BOUNDARY	0.170 / 0.170 mm	Pass
CHALLENGE	0.180 / 0.170 mm	Fail

The hole example assumes a straight cylindrical axis across20 mm, aligned to the datums and allowed clocking. Maximum endpoint radial error is doubled. Real inspection must include size, form, datum simulation and uncertainty; center coordinates alone cannot validate a tilted bore.

11 / Control-plan linkage

An inspection result must lead to a defined reaction.

IDs	Teaching frequency / method	Reaction and evidence
D01	First piece; five consecutive/15 min; micrometer + X-bar/R	Hold on any selected-rule alert or size failure; NCR-001 -> CA-001.
L01	First piece; after setup; height gage/caliper	Independent limits; segregate; inspect stop/setup.
H01/H02	First piece + fixture/tool change; bore + CMM/functional method	Verify size and full axis; hold/NCR on failure.
E01/F01	First piece + process change; face/form method	Recheck datum setup, facing/tooling; hold.
S01	First piece + tool/finish change; profilometer	Check procedure and finishing; hold.
G01	Before adoption + relevant changes; purpose-specific MSA	Physical adequacy required; simulated GR&R alone cannot release method.

The full traceability worksheet maps each ID to a failure mode, candidate cause, prevention/detection control, method, reaction and proposed owner. Ratings are deliberately unscored because no team-approved criteria or production evidence exists.

Sampling frequencies are teaching assumptions. This is not an acceptance-sampling plan with demonstrated outgoing quality or a formal FAI approval. Actual roles would review and sign only after physical evidence exists.

12 / Reproduction and remaining work

What is finished, what is conditional, and what still requires access.

Deliverable	Status
Python SPC and capability	Executed, with exported raw records, subgroup calculations, all alarms and diagnostics.
NCR / RCA / control linkage	Complete training documentation; effectiveness remains open by the reported criterion.
Drawing / first-piece record	Authored learning addendum and simulated examples; native Onshape revision not performed.
Minitab comparison	Prepared inputs, exact options and Python reference; software execution pending sign-in/license.
Phase 1	Preserved repaired simulation, GR&R, inspection logic, Excel dashboard and original drawing.

Rebuild

Install requirements.txt. Run python scripts/build_quality_package.py, then python scripts/build_spc_notebook.py. Execute python -m unittest discover -s tests -v. CSVs and JSON retain full precision; rounded values on pages are for readability.

Sources

NIST: Shewhart X-bar and R charts (itl.nist.gov/div898/handbook/pmc/section3/pmc311.htm); process capability ([section1/pmc16.htm](https://itl.nist.gov/div898/handbook/pmc/section1/pmc16.htm)). Minitab official documentation: Xbar-R special-cause tests and normal-capability methods. ASME Y14.5 scope and Tec-Ease position interpretation. Full links and option details appear in the Phase2, inspection and Minitab README files. Accessed14 Sep2026.

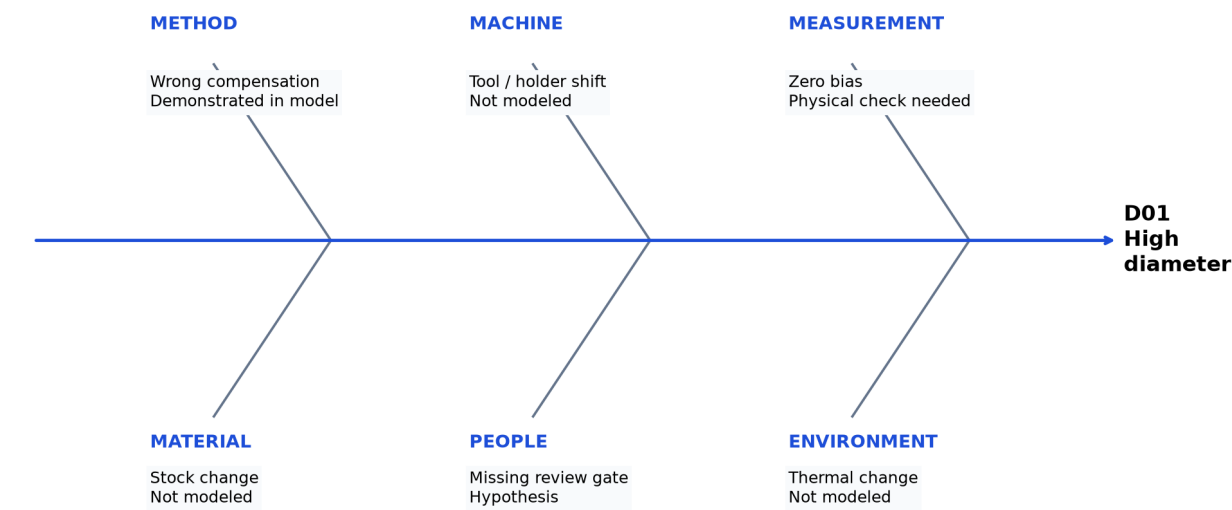
Resume boundary

This release supports discussion of simulated SPC, measurement interpretation, capability conditions, inspection planning and corrective-action documentation. Add Minitab only after completing genuine software runs; keep GD&T qualified as fundamentals / drawing interpretation.

13 / Root-cause map

Hypotheses remain visible beside the demonstrated model cause.

NCR-001 | Candidate causes and evidence status



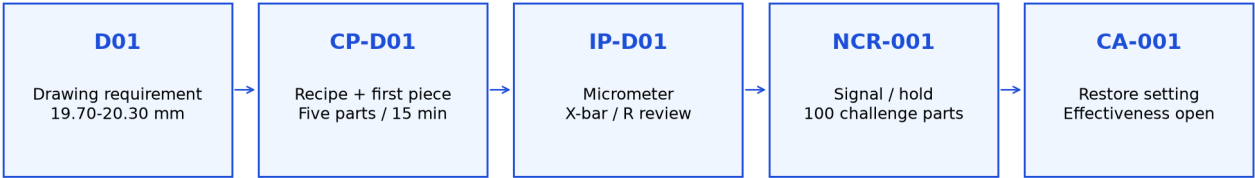
SIMULATED RCA | Only the programmed compensation change is demonstrated.

The method branch identifies the programmed compensation error. The other branches identify evidence that a real investigation would collect. They are not claims that alternative physical causes have been ruled out.

14 / Traceability at a glance

From a size requirement to a documented effectiveness decision.

One requirement connects the quality documents



Verification: 0 / 200 diameter rejects, with a retained trend alert at subgroup 122.

SIMULATED QUALITY PLANNING | Proposed methods and ownership

Follow D01 through the requirement, prevention/detection control, inspection method, containment record and corrective action. The open effectiveness decision is backed by subgroup122 in the alarm log.